

Answer: A

12 Determining the phase angle at $d = 5.0$ cm,

$$y = y_0 \sin \theta$$

$$1.5 = 3.0 \sin \theta$$

$$\theta = \sin^{-1}\left(\frac{1.5}{3.0}\right) = \frac{\pi}{6}$$

Since $d = 5.0$ cm is in the second quarter of the second wavelength,

$$\text{phase angle } \phi = 3\pi - \frac{\pi}{6} = 2\frac{5}{6}\pi$$

Hence,

$$\frac{\Delta\phi}{2\pi} = \frac{\Delta d}{\lambda}$$

$$\frac{2\frac{5}{6}\pi}{2\pi} = \frac{5.0}{\lambda}$$

$$\lambda = 3.5 \text{ cm}$$

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